

THIS PAGE LEFT INTENTIONALLY BLANK

THIS PAGE LEFT INTENTIONALLY BLANK

REAR OF TGS:

The following must be applied when installing the attached TGS for Asphalt works to be completed on Scott Street; Harristown NSW (see [Figure E-1.](#))

STEP	ACTION
1	Depart the meeting point
2/3/4	Install R4-212 (40), 'End Road Work/speed reinstatement' and install R4-212 (60) as the first sign when initially leaving work area.
5	Use the existing road network to turn where safe to do so
6/7/8/9/10/11	Then place approach signage signs, including the portable boom barrier in upright position (TC (b) to remain with boom barrier).
12/13	Then install R4-212 (40) turn left into Boyce Street, install 'End Road Work/speed reinstatement'
14	Use the existing road network to turn where safe to do so
15/16/17/18/19	Then place approach signage signs, including the portable boom barrier in upright position (TC (b) to remain with boom barrier).
20/21	Then turn left Scott St install R4-212 (40), then turn left Avery Street, install 'End Road Work/speed reinstatement'
22	Use the existing road network to turn where safe to do so
23/24/25/26/27	Then place approach signage signs, including the portable boom barrier in upright position (TC (b) to remain with boom barrier).
28/29/30	Then turn left Twice Road, install 'End Road Work/speed reinstatement' and install R4-212 (60)
31	Use the existing road network to turn where safe to do so
32/33/34/35/36/37	Then place approach signage signs, including the portable boom barrier in upright position (TC (b) to remain with boom barrier).
38/39	Then install both R4-212 (40).
40	Return to meeting point and confirm readiness with TCs

Appendix D – Work type layout examples

INFORMATIONAL APPENDIX

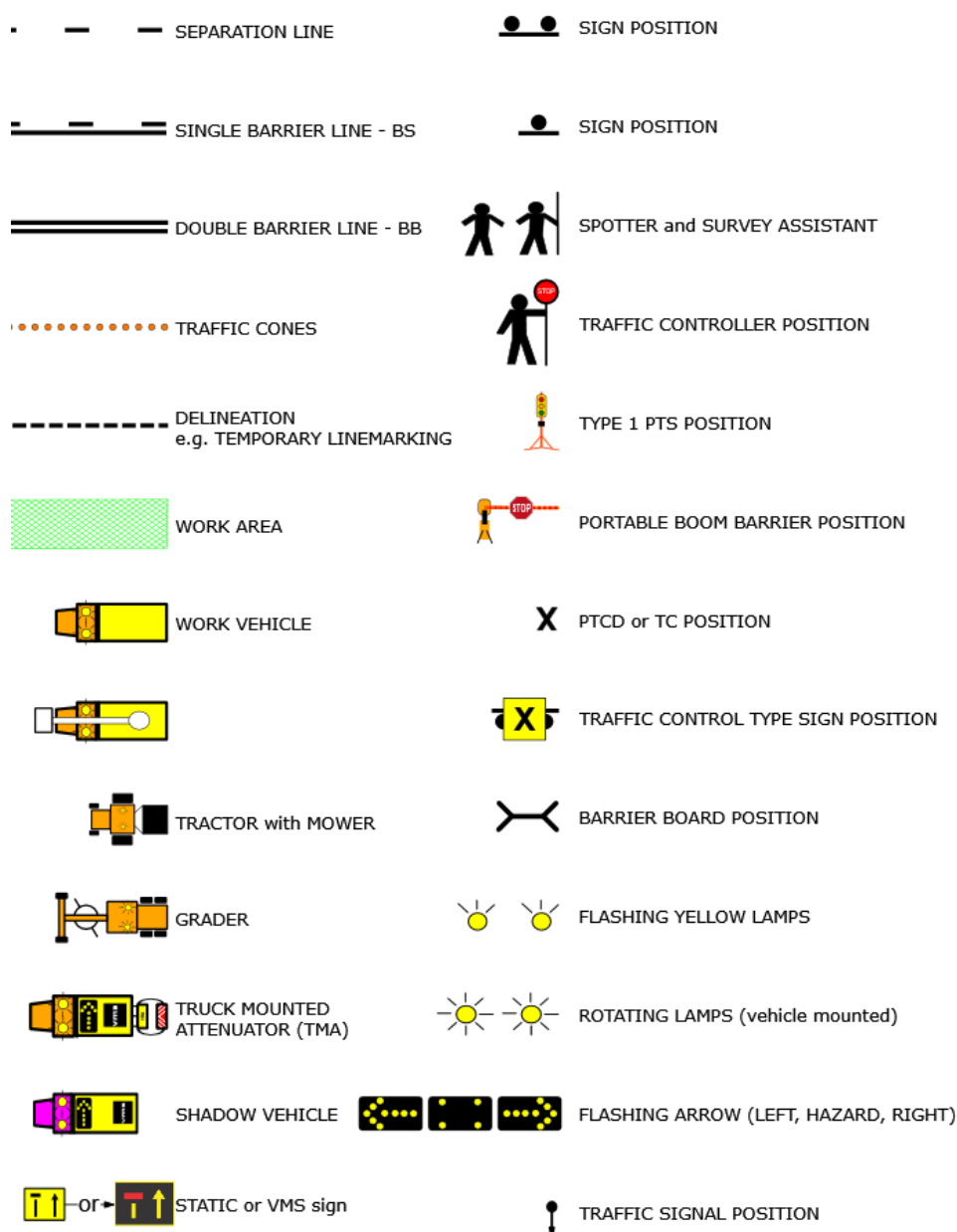
D.1 Scope

This Appendix provides various examples of work type layouts that may be referenced when developing traffic guidance schemes (TGSs). The requirements for preparing TGSs are provided in [Section 7 Traffic guidance schemes \(TGS\)](#).

Note: These examples reflect the minimum requirements, additional signage and advance warnings based on a risk assessment may be required.

D.2 Standard symbols

The standard symbols used in the work type layouts are provided below.

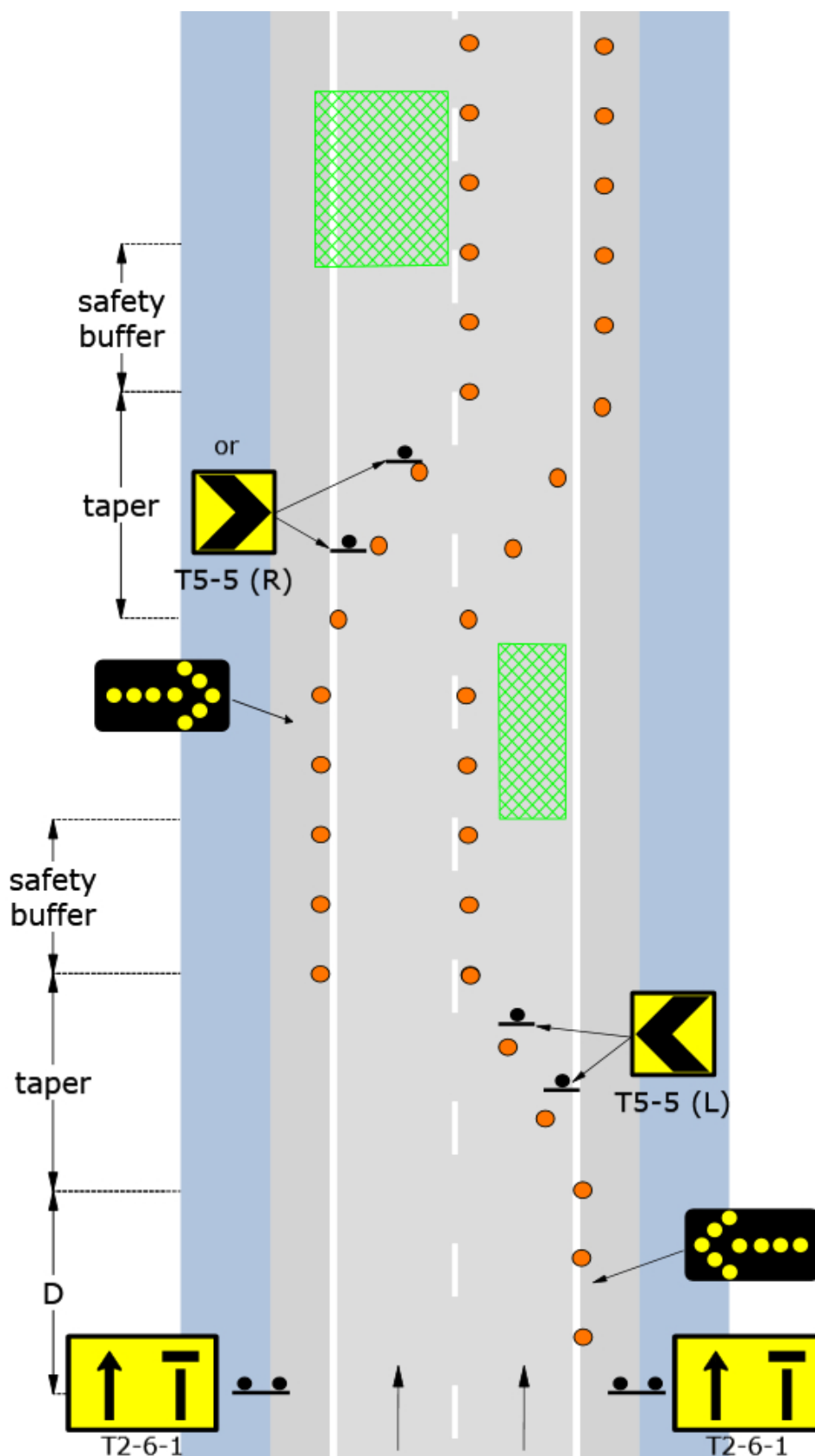


D.3 Table of examples

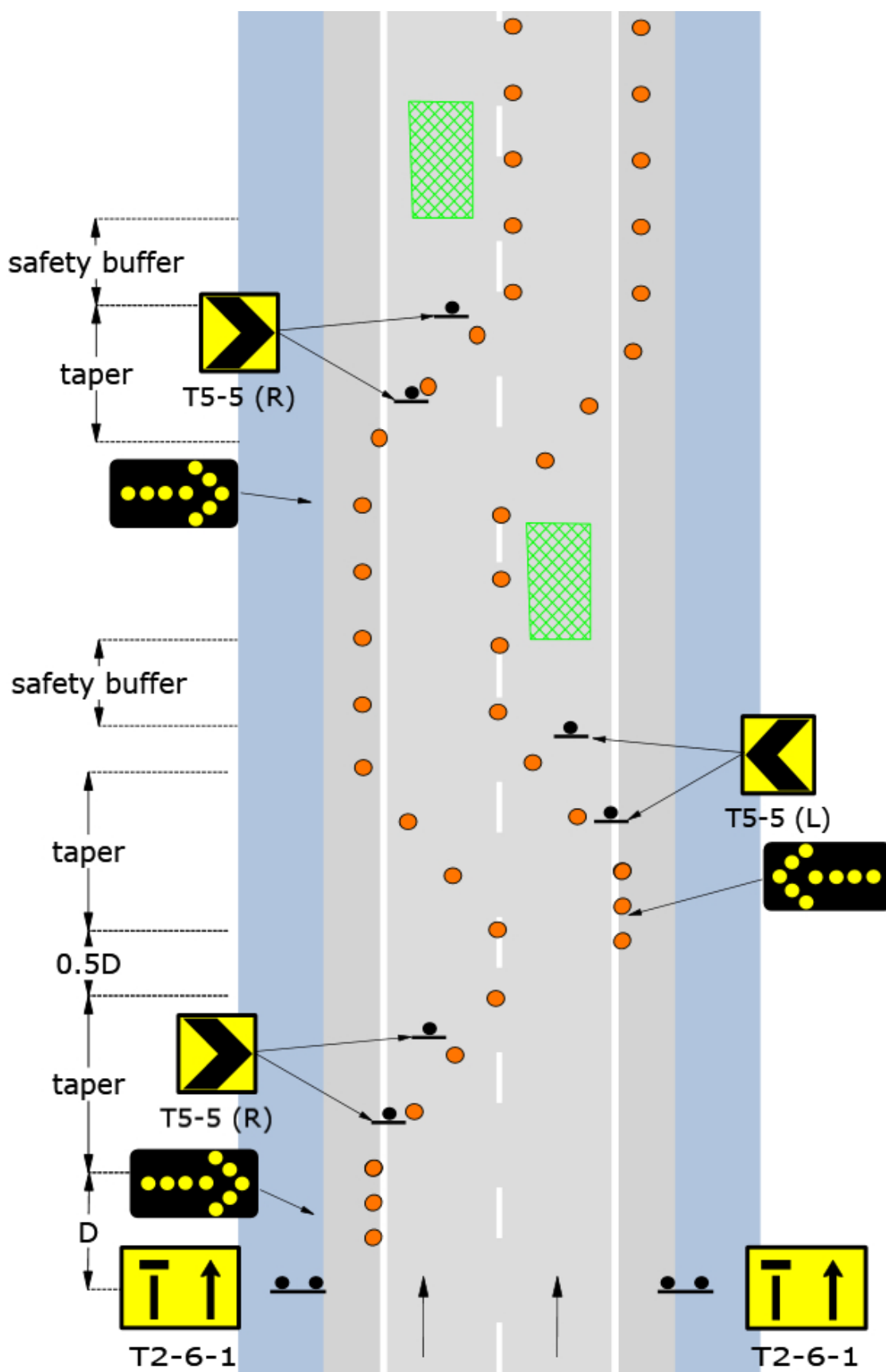
Reference	Work Type	Description
D.4.1	Static Work	Lane closure with chicane - 2 lane / divided (formerly TCP 94) – right to left merge
D.4.2	Static Work	Lane closure with double chicane – 2 lane/divided (formerly TCP 102) – left to right merge
D.4.3	Static Work	Dual lane closure - 3 lane / divided (formerly TCP 100)
D.4.4	Static Work	Dual lane closure for work in fast lane - 3 lane / 2 way (formerly TCP 90)
D.4.5	Static Work	Use of truck-mounted attenuator to support installation of lane closure
D.4.6	Static Work	Road closure using multi message signs
D.4.7	Static Work	Access to depot, stockpile, quarry, gravel pit etc. all roads (formerly TCP 195)
D.4.8	Static Work	Bitumen sealing - 2 lane / 2 way (formerly TCP 95)
D.4.9	Static Work	Long term speed reduction (formerly TCP 57)
D.4.10	Static Work	Use of temporary portable rumble strips (shuttle flow)
D.4.11	Static Work	Use of temporary portable rumble strips (merge lane closure)
D.4.12	Dynamic Work	Continuous work - use of truck -mounted attenuator
D.4.13	Dynamic Work	Continuous work - multi lane (formerly TCP 62)
D.4.14	Dynamic Work	Continuous work in traffic - 3 lane / 2 way work (formerly TCP 70)
D.4.15	Dynamic Work	Continuous work - clear of traffic (formerly TCP 40)
D.4.16	Dynamic Work	Continuous work, grading - 2 lane / 2 way (formerly TCP 105)
D.4.17	Dynamic Work	Continuous work middle lane closed 3 lane / divided (formerly TCP 107)
D.4.18	Dynamic Work	Frequently changing work - 2 lane / 2 way for ADT < 1,500 (formerly TCP 77)
D.4.19	Dynamic Work	Frequently changing work in shoulder (formerly TCP 831)
D.4.20	Dynamic Work	Intermittent work, surveying (formerly TCP 459)

D.4 Work type examples

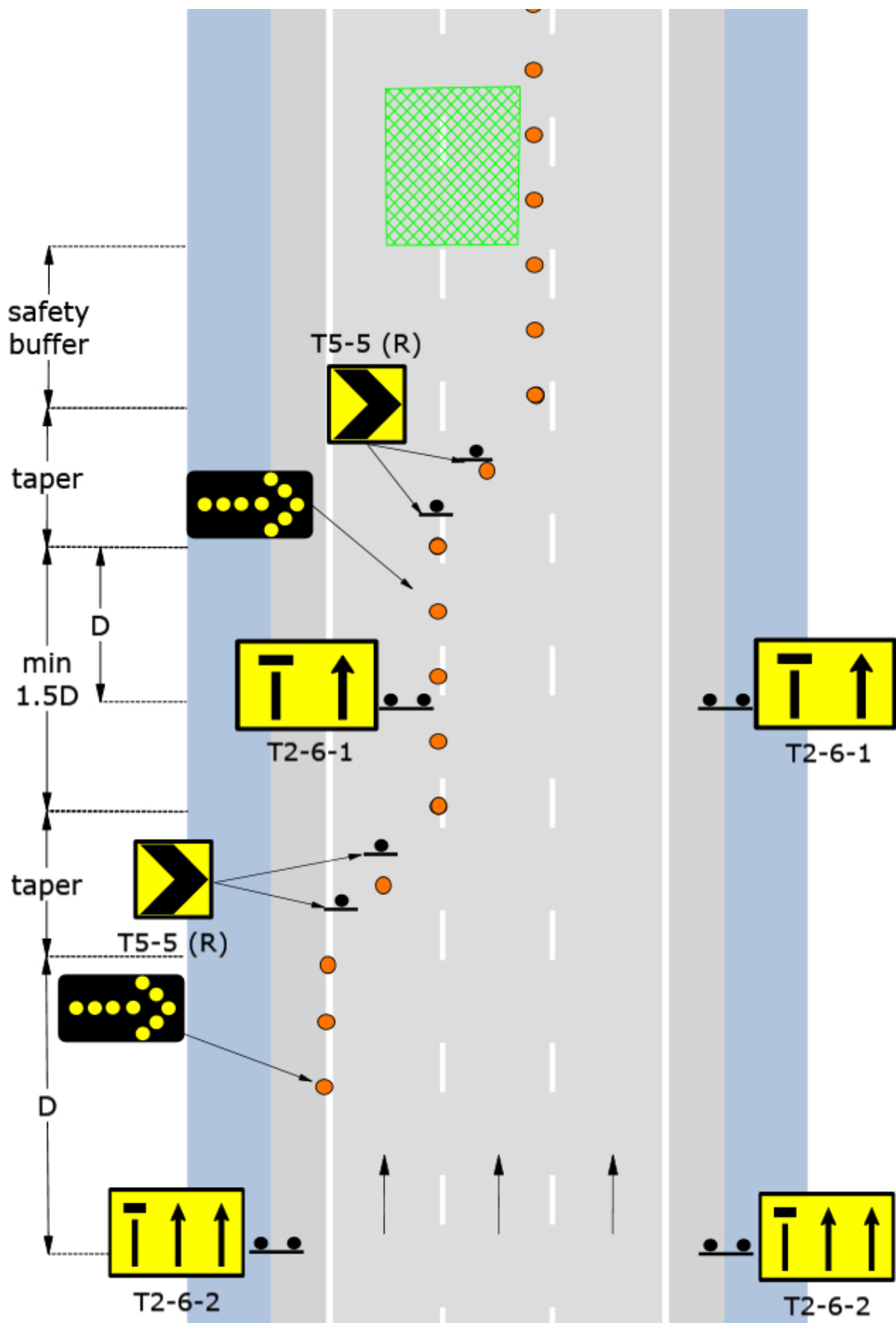
D.4.1 Static: Lane closure with chicane - 2 lane / divided (formerly TCP 94) – right to left merge



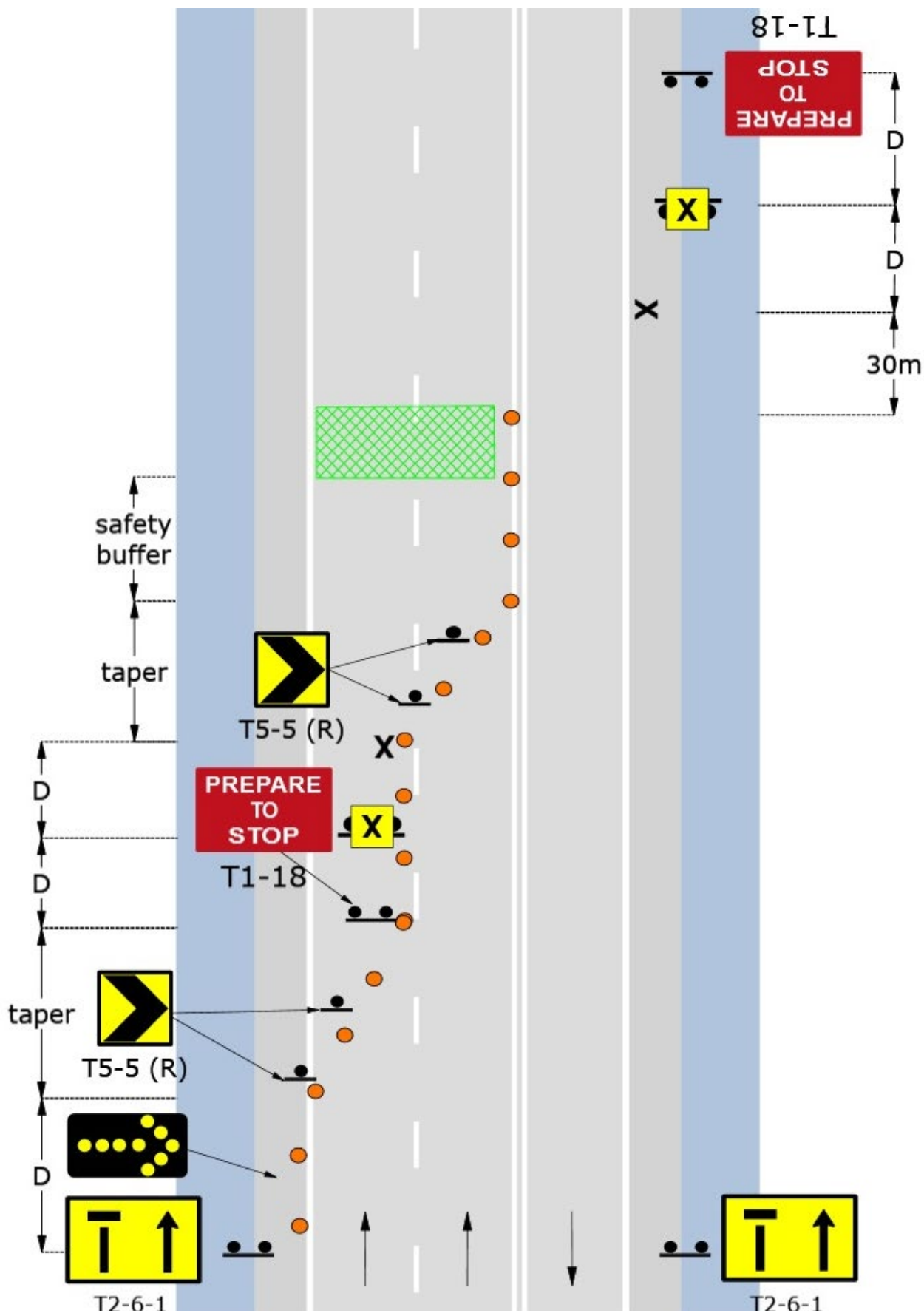
D.4.2 Static: Lane closure with double chicane – 2 lane/divided (formerly TCP 102) – left to right merge



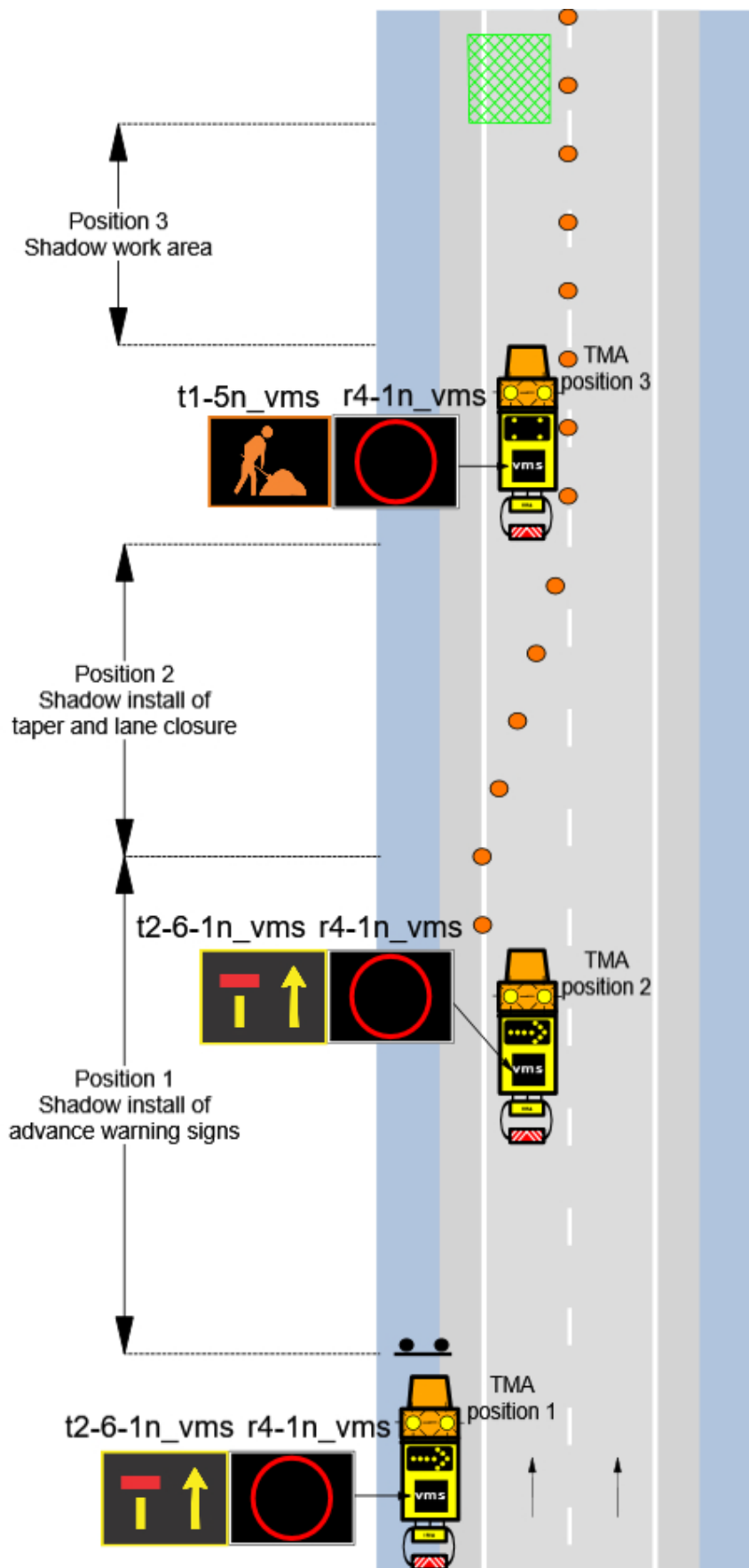
D.4.3 Static: Dual lane closure - 3 lane / divided (formerly TCP 100)



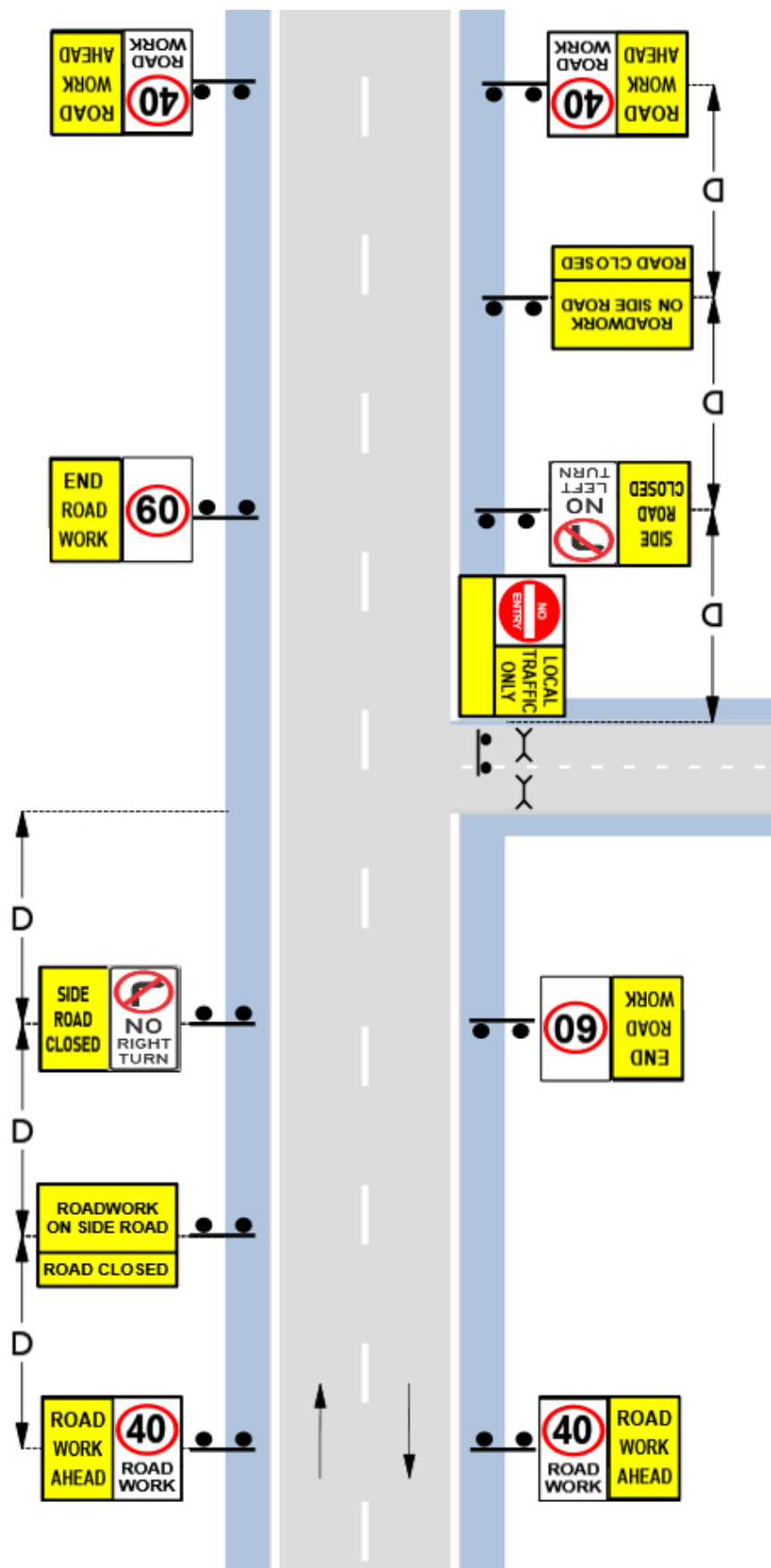
D.4.4 Static: Dual lane closure for work in fast lane - 3 lane / 2 way (formerly TCP 90)



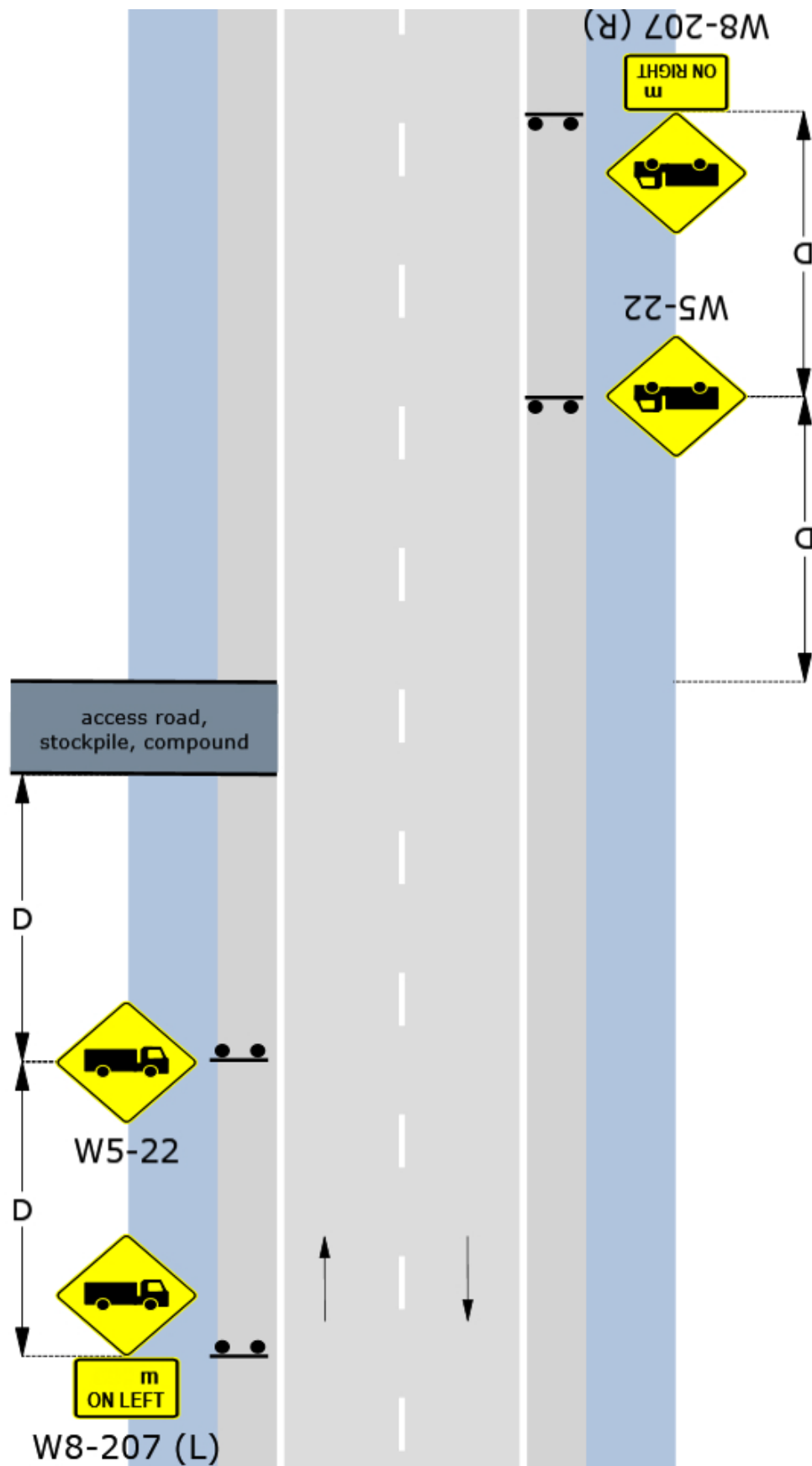
D.4.5 Static: Use of truck-mounted attenuator to support installation of lane closure



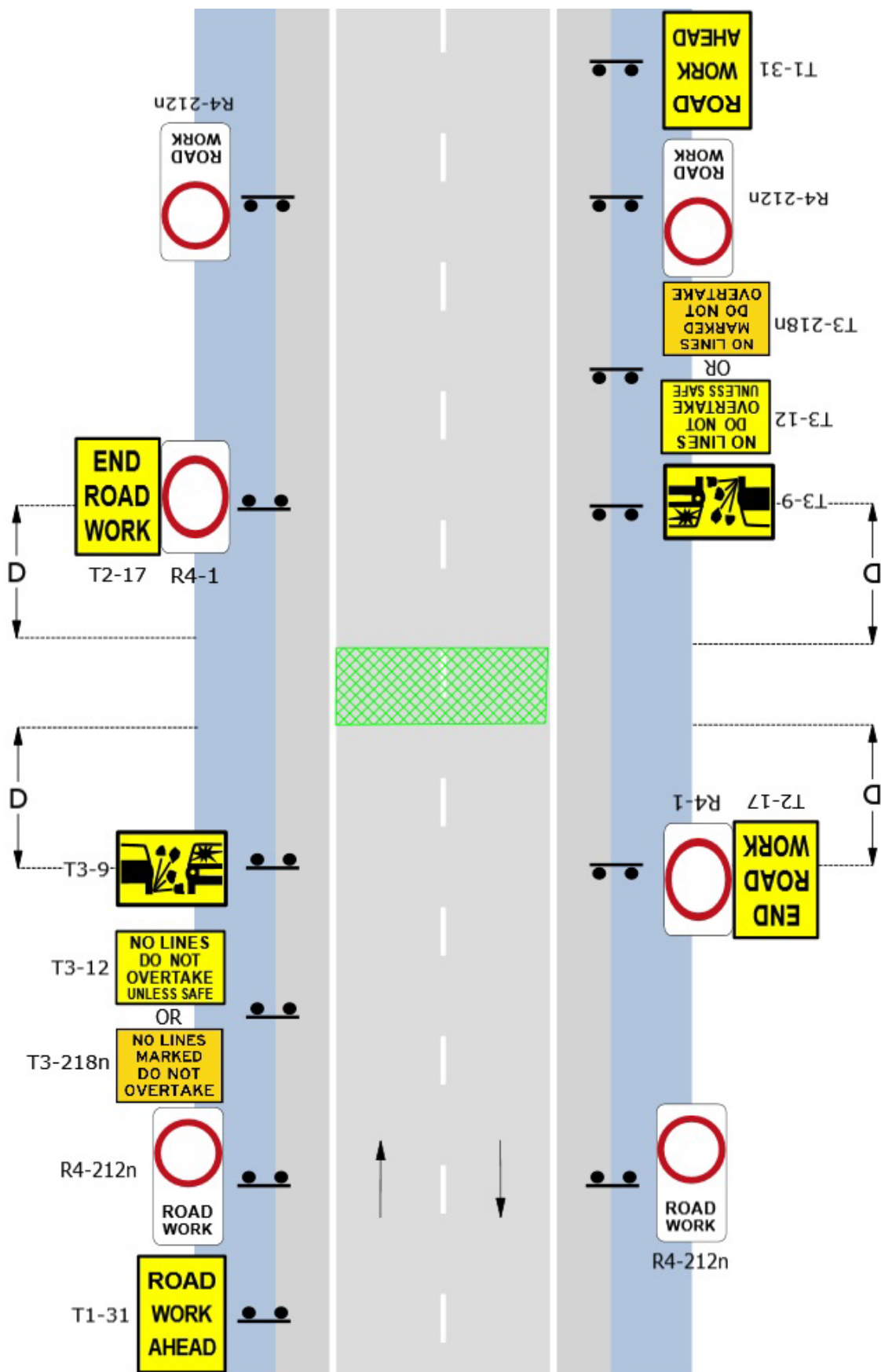
D.4.6 Static: Road closure using multi message signs



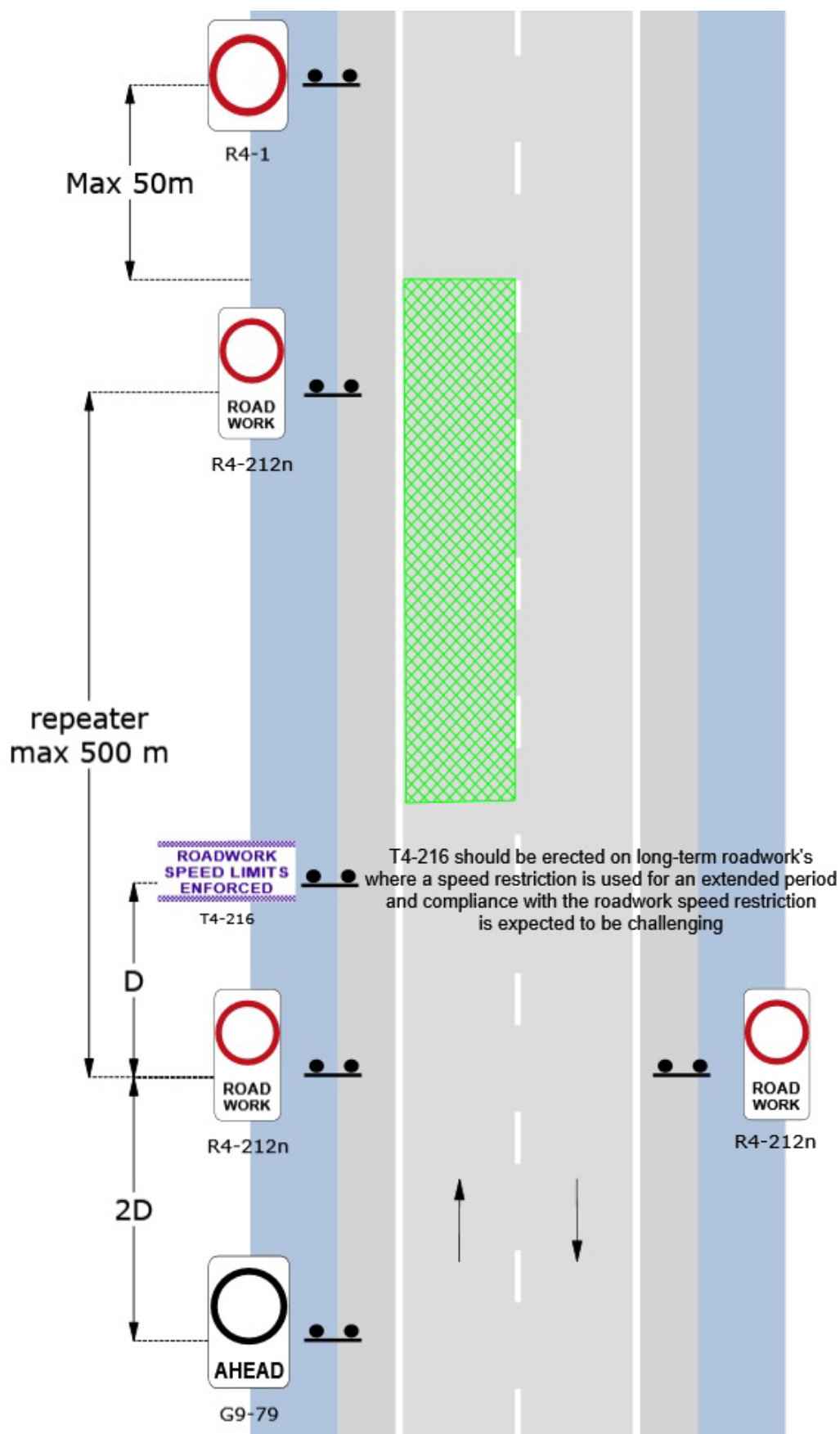
D.4.7 Static: Access to depot, stockpile, quarry, gravel pit etc. all roads (formerly TCP 195)

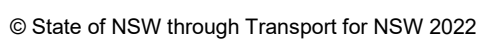


D.4.8 Static: Bitumen sealing - 2 lane / 2 way (formerly TCP 95)

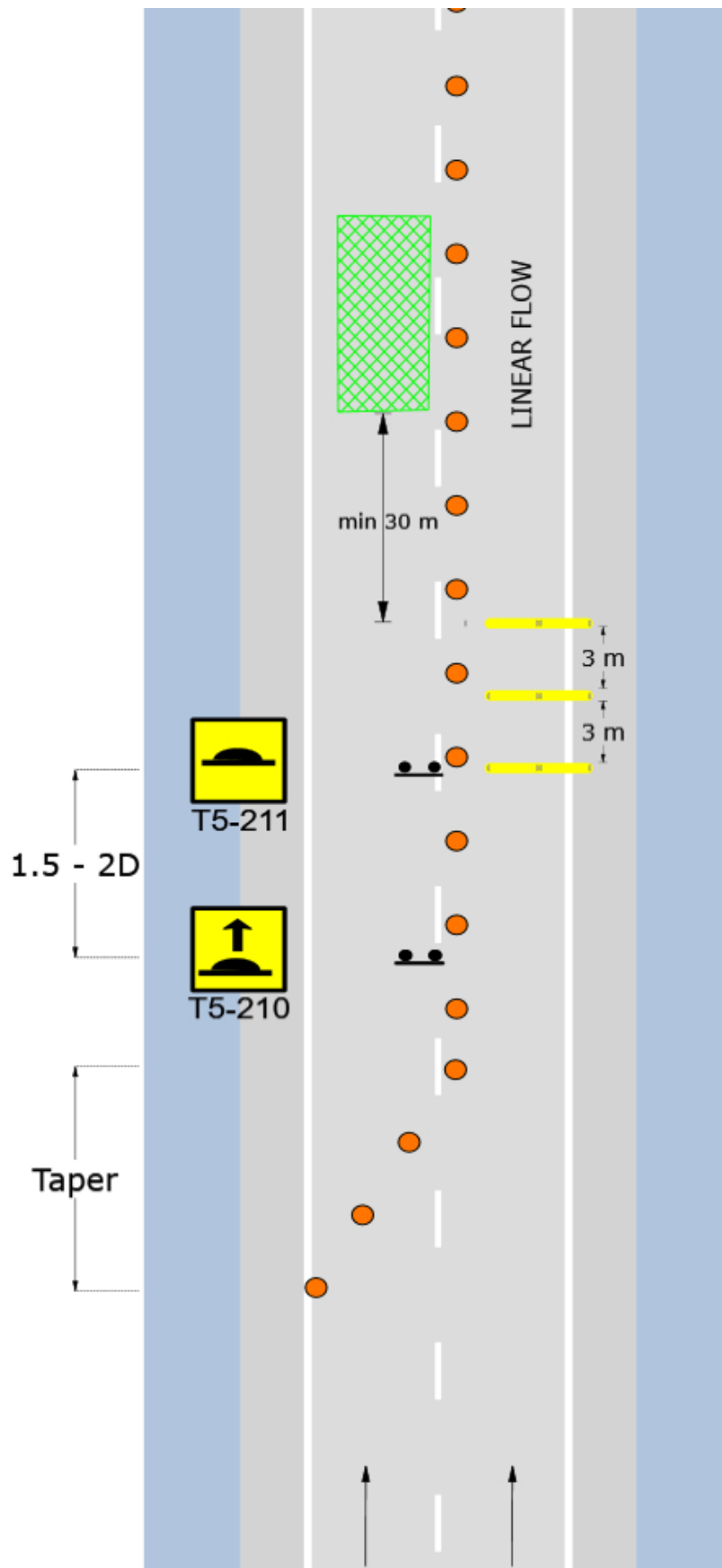


D.4.9 Static: Long Term speed reduction (formerly TCP 57)

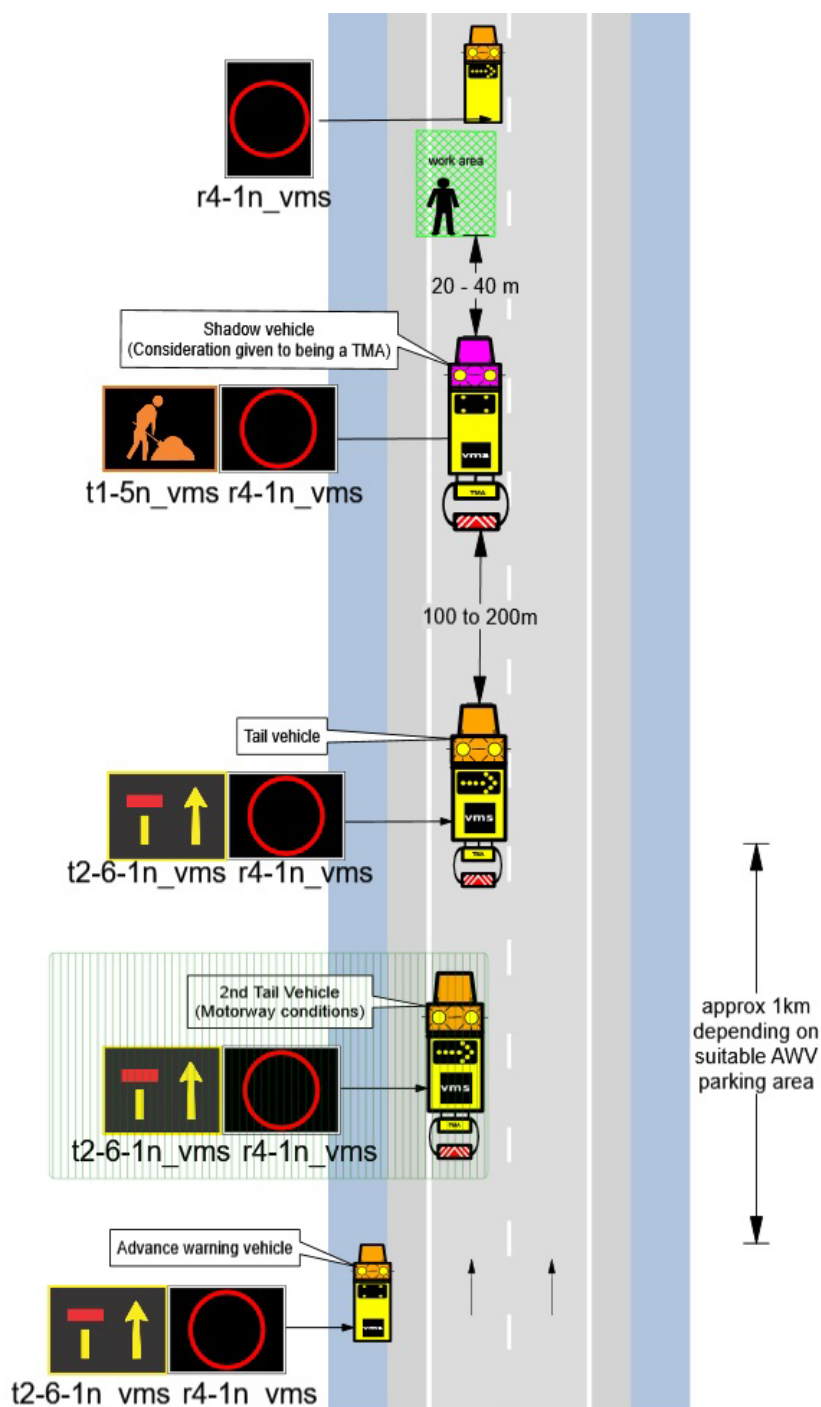




D.4.11 Static: Use of temporary portable rumble strips (merge lane closure)



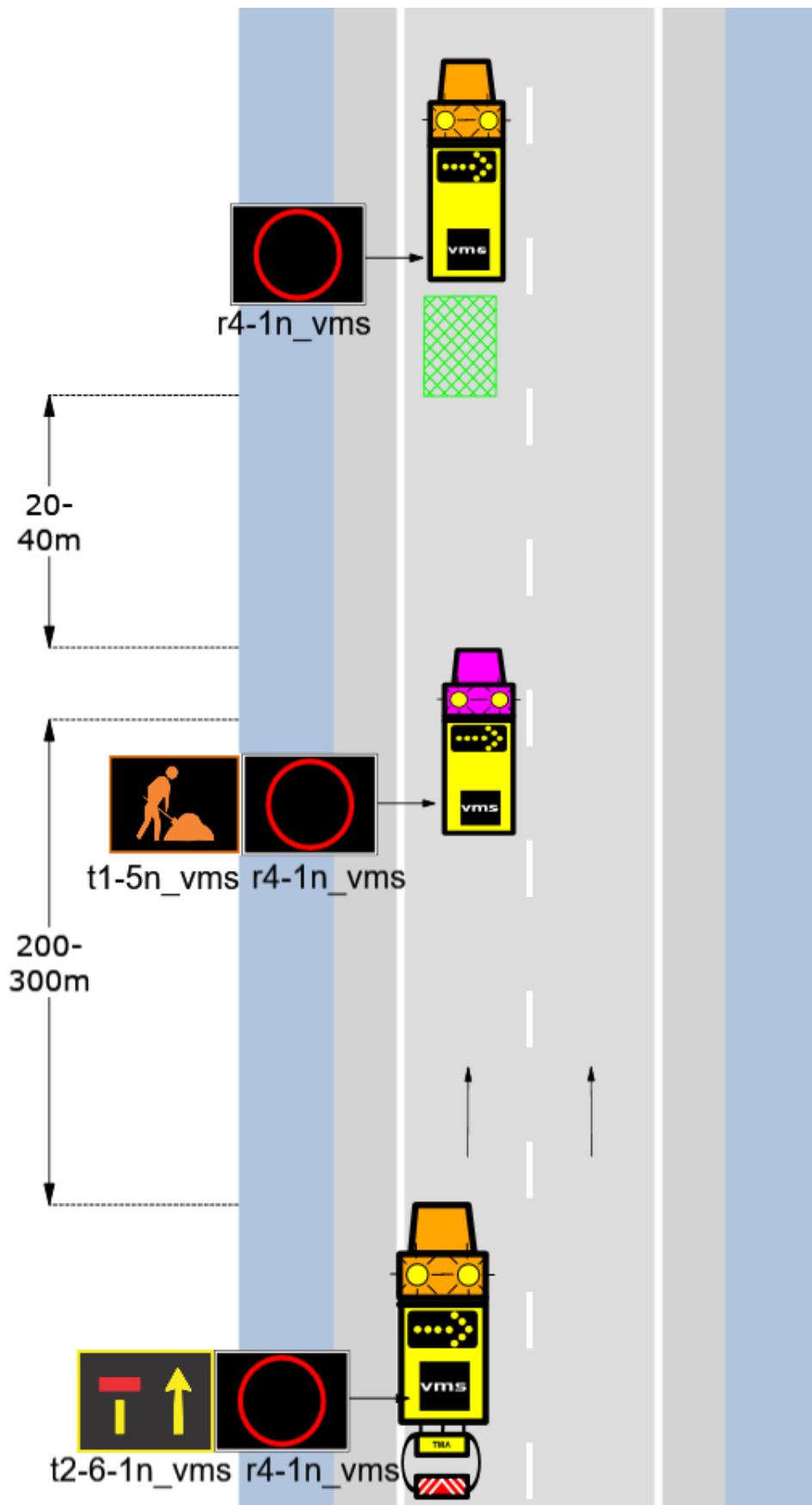
D.4.12 Dynamic: Continuous work - use of truck -mounted attenuator



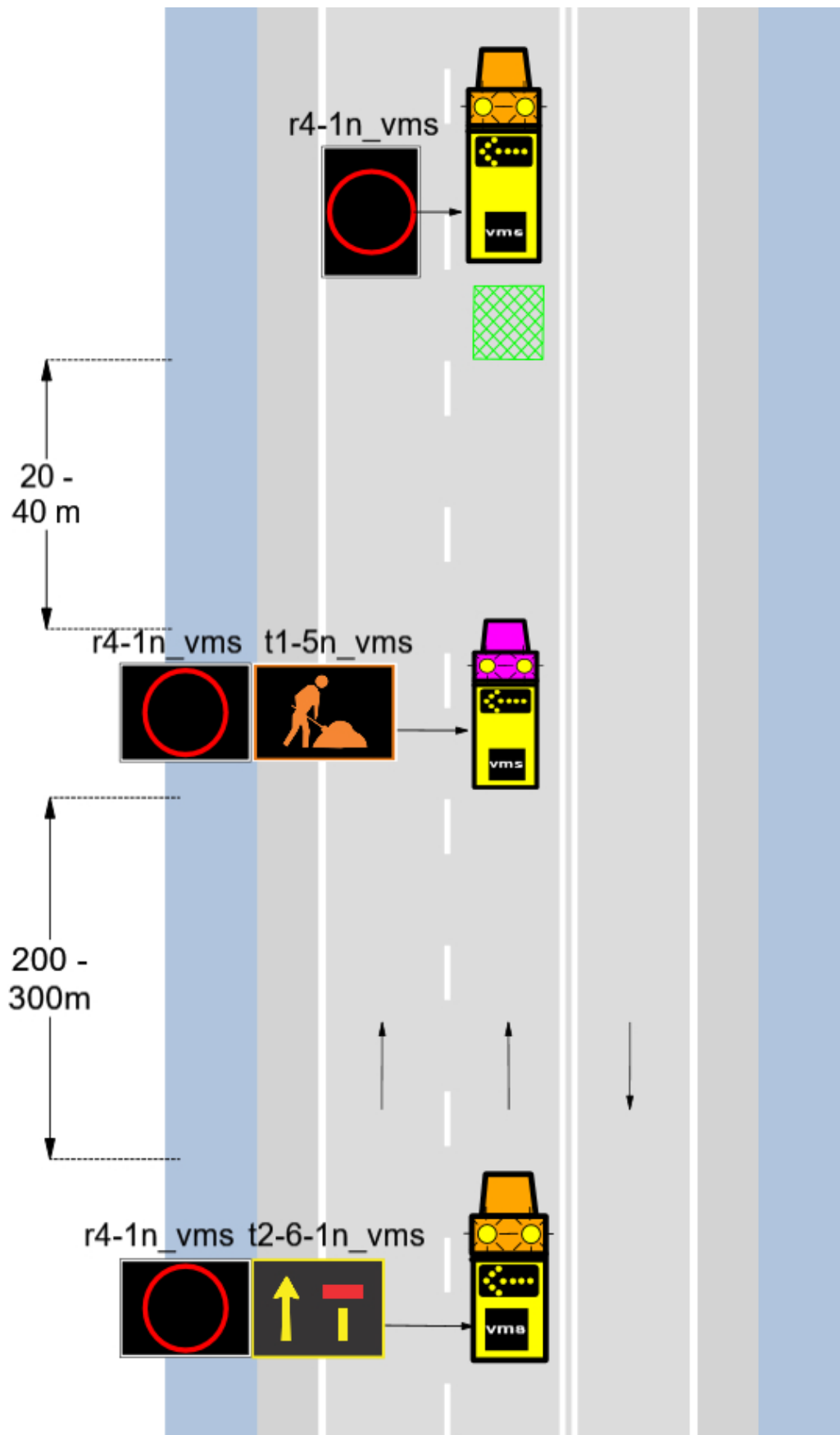
Notes to Figure:

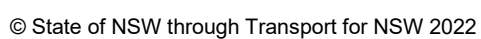
1. Advance warning vehicle positioned greater than 500 mm inside the edge line and should include a VMS.
2. The TMA should be the first vehicle road users encounter in the lane, unless in motorway conditions, in which case the second tail vehicle will be encountered first, which straddles the edge line.
3. The tail vehicle TMA should be positioned within the travel lane.
4. The shadow vehicle should be positioned relative to worker on foot clearance requirements.
5. Vehicle spacing should be maintained; minimum sight distance to mounted warning devices should be 250 m.
6. Regulatory signs (e.g. speed signs) on a VMS must display continuously, and not alternate with another sign.

D.4.13 Dynamic: Continuous work - multi lane (formerly TCP 62)

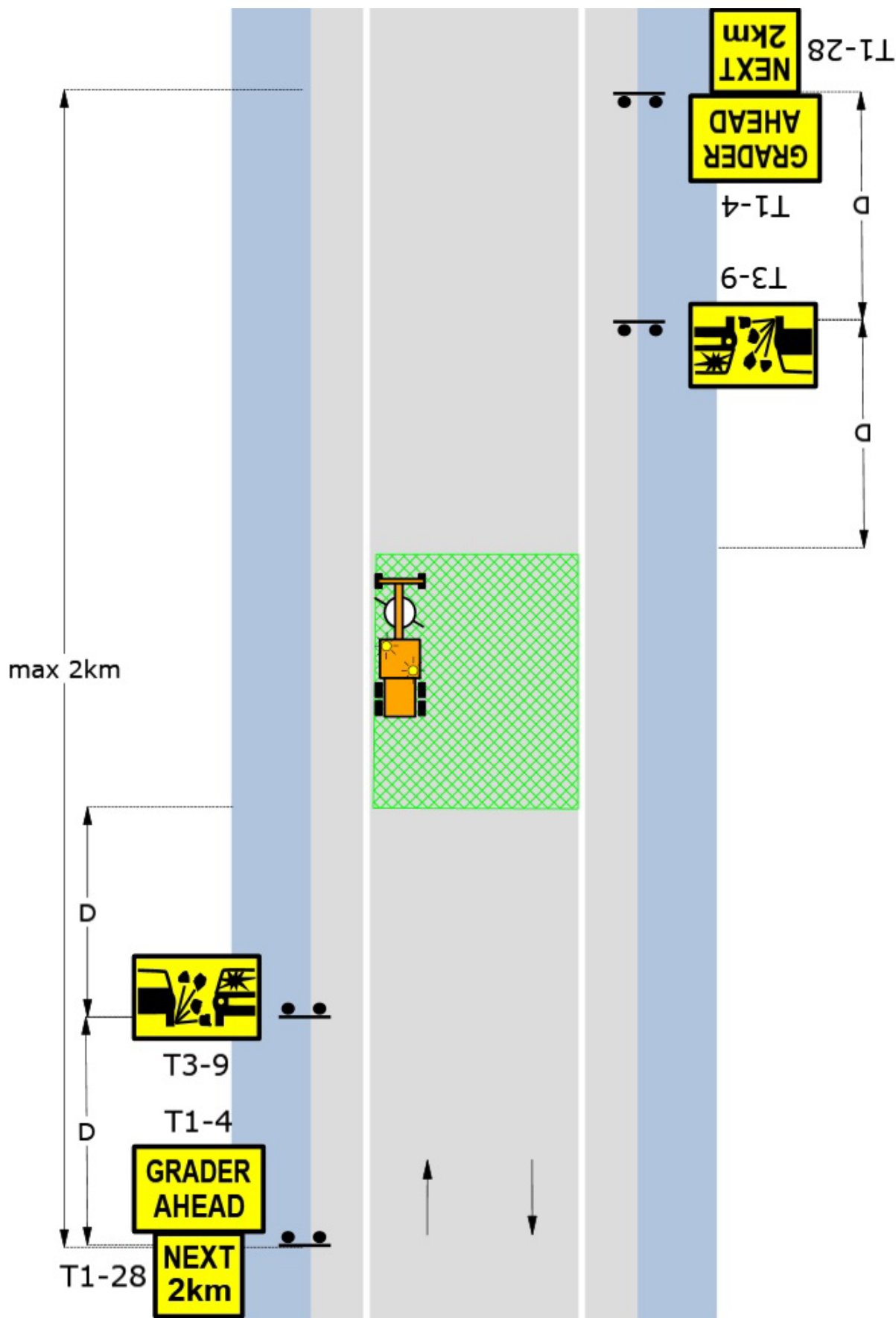


D.4.14 Dynamic: Continuous work in traffic - 3 lane / 2 way work (formerly TCP 70)

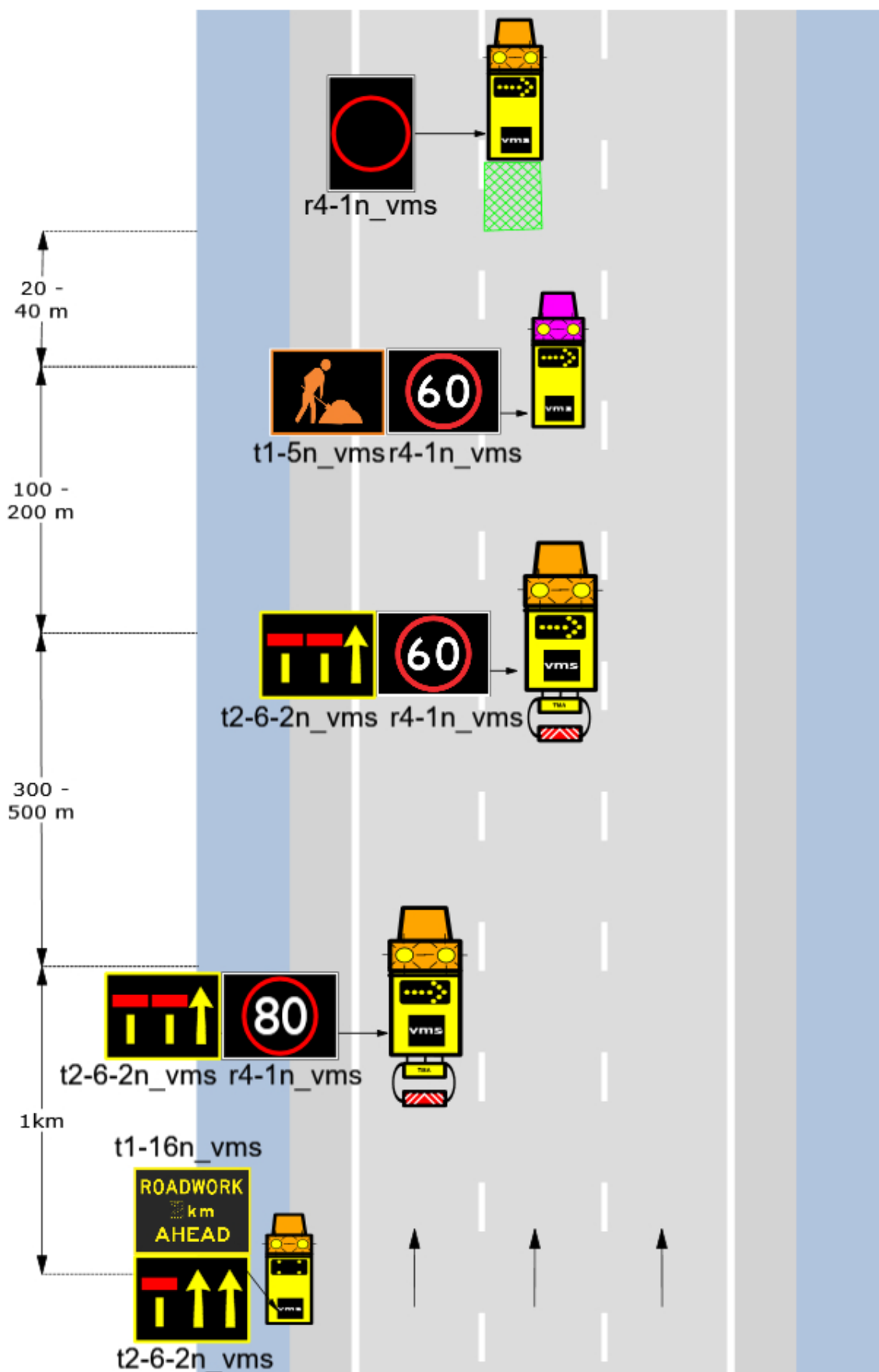




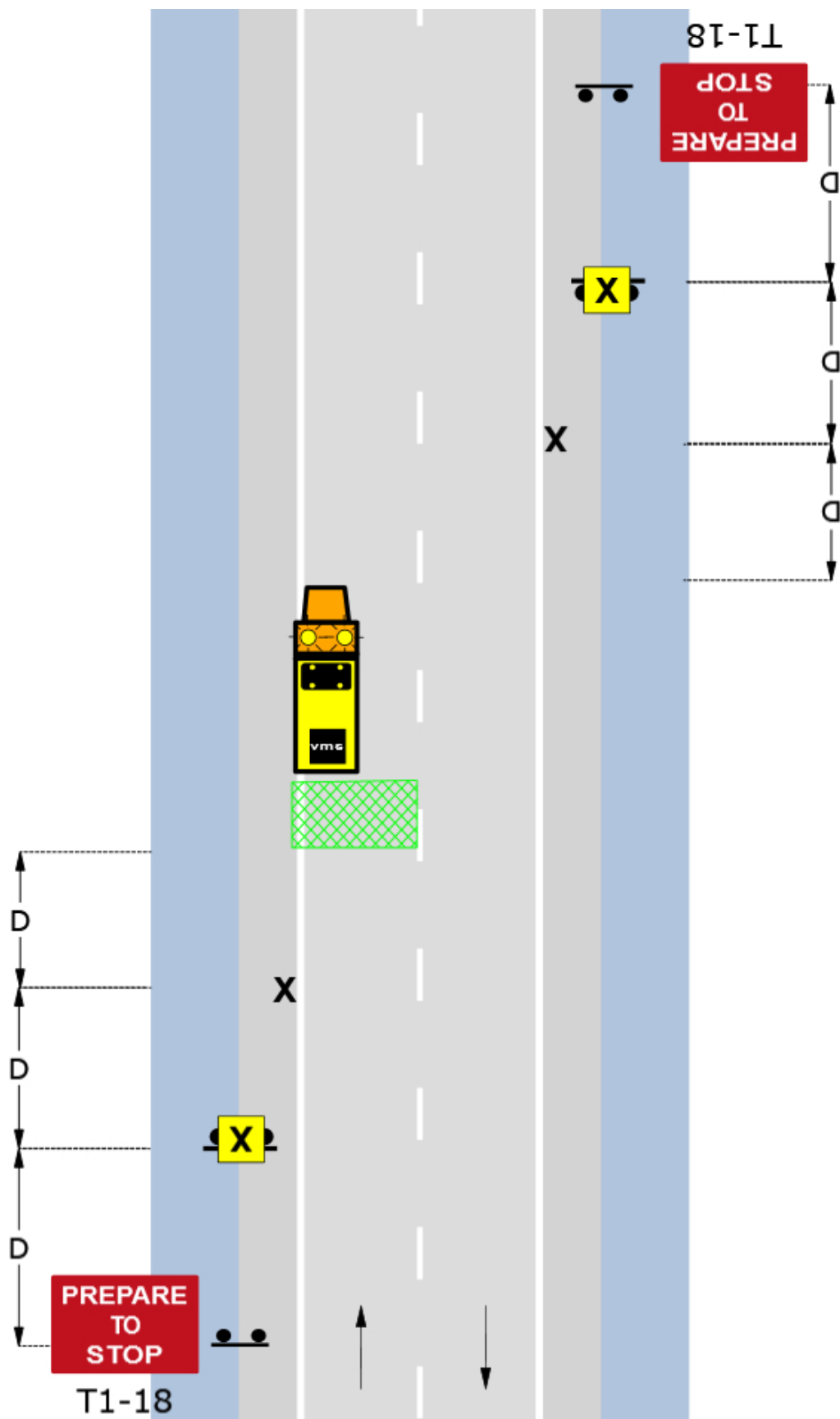
D.4.16 Dynamic: Continuous work, grading - 2 lane / 2 way (formerly TCP 105)



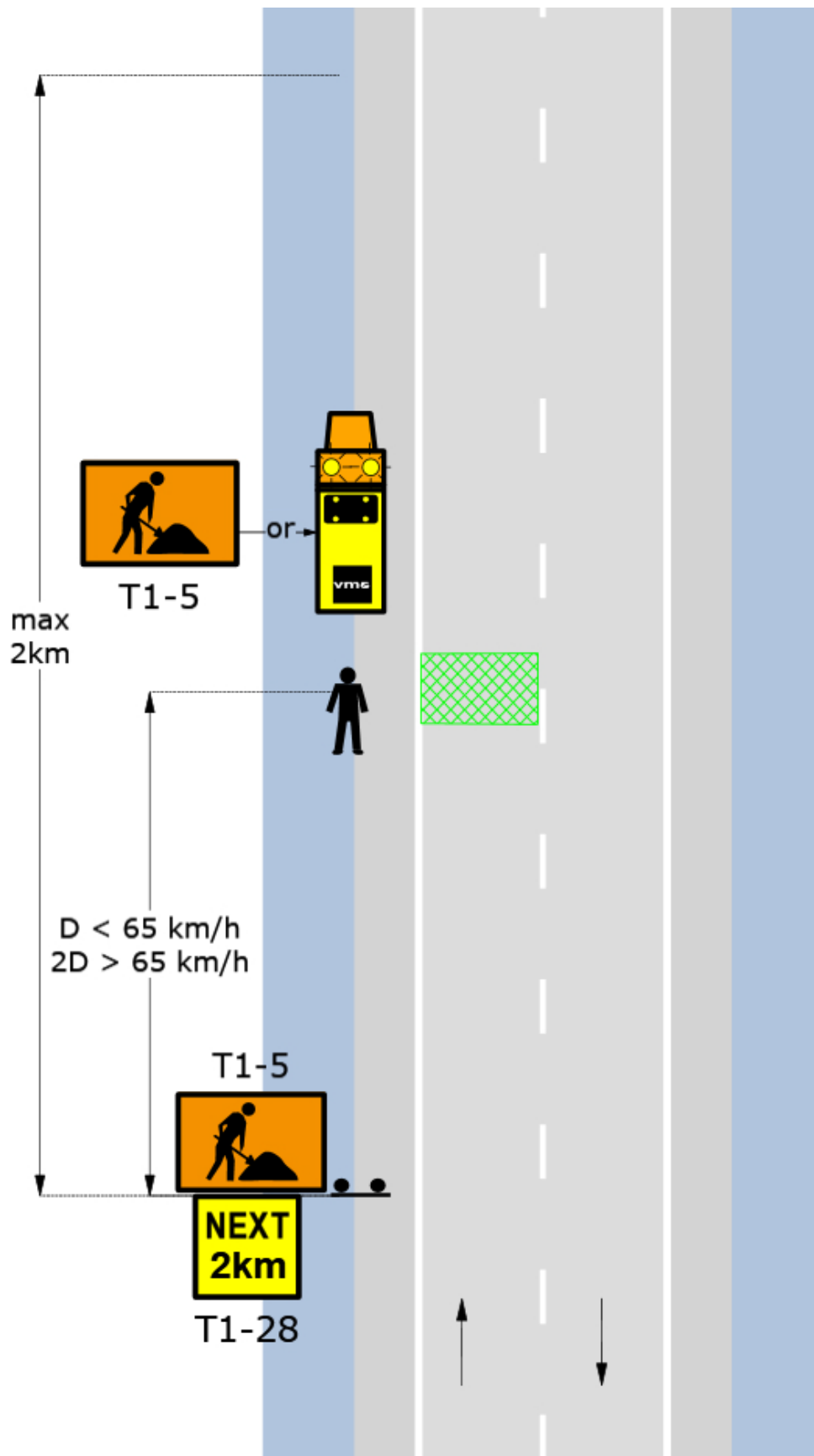
D.4.17 Dynamic: Continuous work middle lane closed 3 lane / divided (formerly TCP 107)



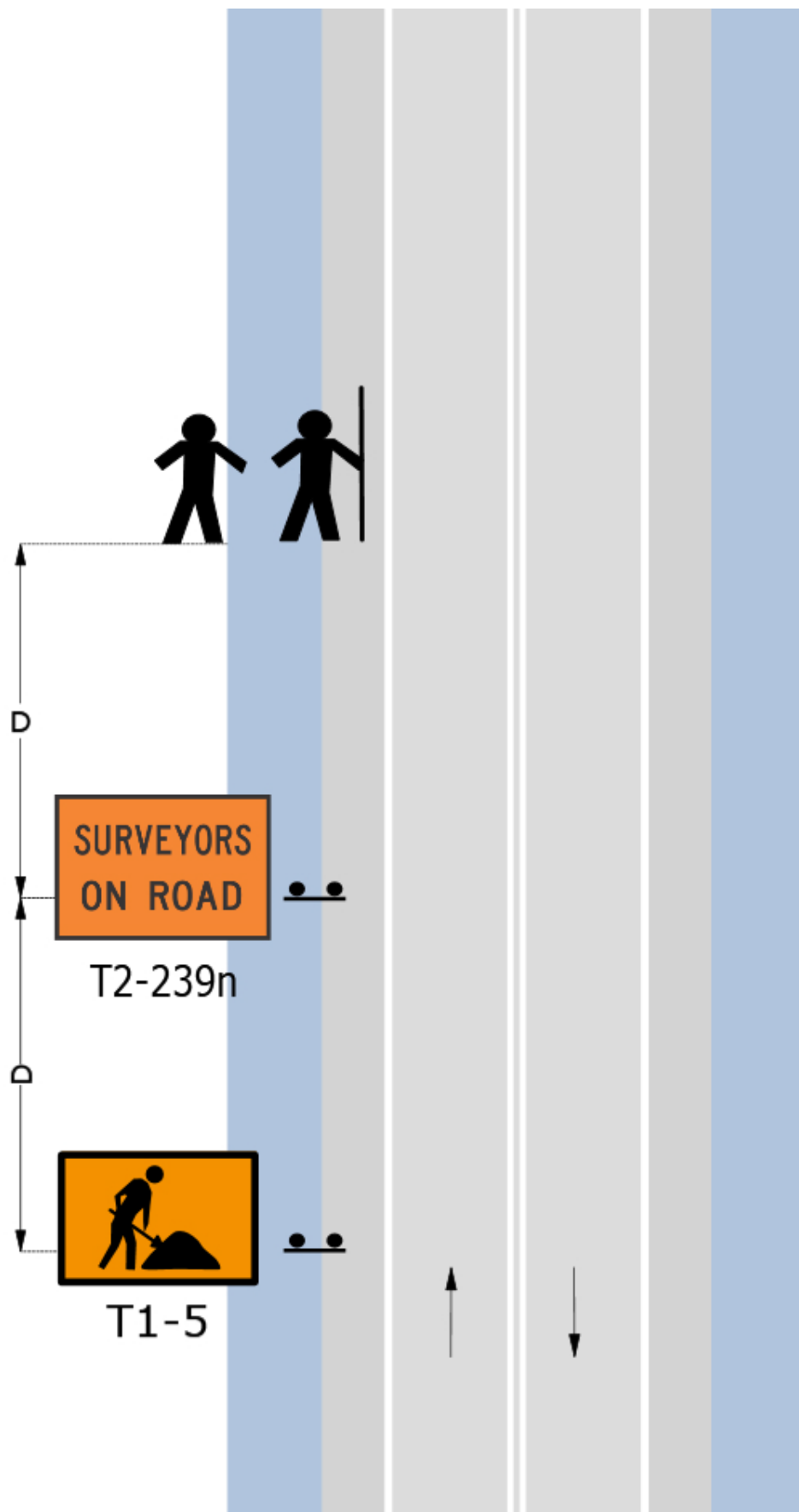
D.4.18 Dynamic: Frequently changing work - 2 lane / 2 way for ADT less than 1,500 (formerly TCP 77)



D.4.19 Dynamic: Frequently changing work in shoulder (formerly TCP 831)



D.4.20 Dynamic: Intermittent work, surveying (formerly TCP 459)



Appendix E – Inspection checklists and tools

INFORMATIONAL APPENDIX

E.1 Scope

This Appendix provides a number of checklists that may be used by project teams in relation to inspections. While every attempt has been made to align these checklists with the requirements of this Technical Manual, the checklists are examples only. Completion of these checklists might not necessarily fulfil all of the relevant requirements of this Technical Manual. And thus, pertaining to each work site, it remains the obligation of project teams to ensure all relevant requirements of this Technical Manual have been met regardless of the content of these example checklists.

E.2 TGS verification checklist

TGS Verification must be undertaken after selecting or designing a TGS as a confirmation of appropriateness prior to approval for use. A PWZTMP or ITCP qualified person must undertake this verification.

Completed by:			
Name:		Signature:	
Qualification			
TGS details:			
TMP Reference:		TGS Reference:	
Date:		Review type	☐ Site Inspection ☐ Desktop review
Sources used for desktop review			
Site details			
Street name:		Confirmed posted speed limits:	
Street name:		Confirmed posted speed limits:	
Street name:		Confirmed posted speed limits:	
List unique site specific Hazards / Risks identified on site			
<i>E.g. utilities, infrastructure, vegetation, schools,</i>			

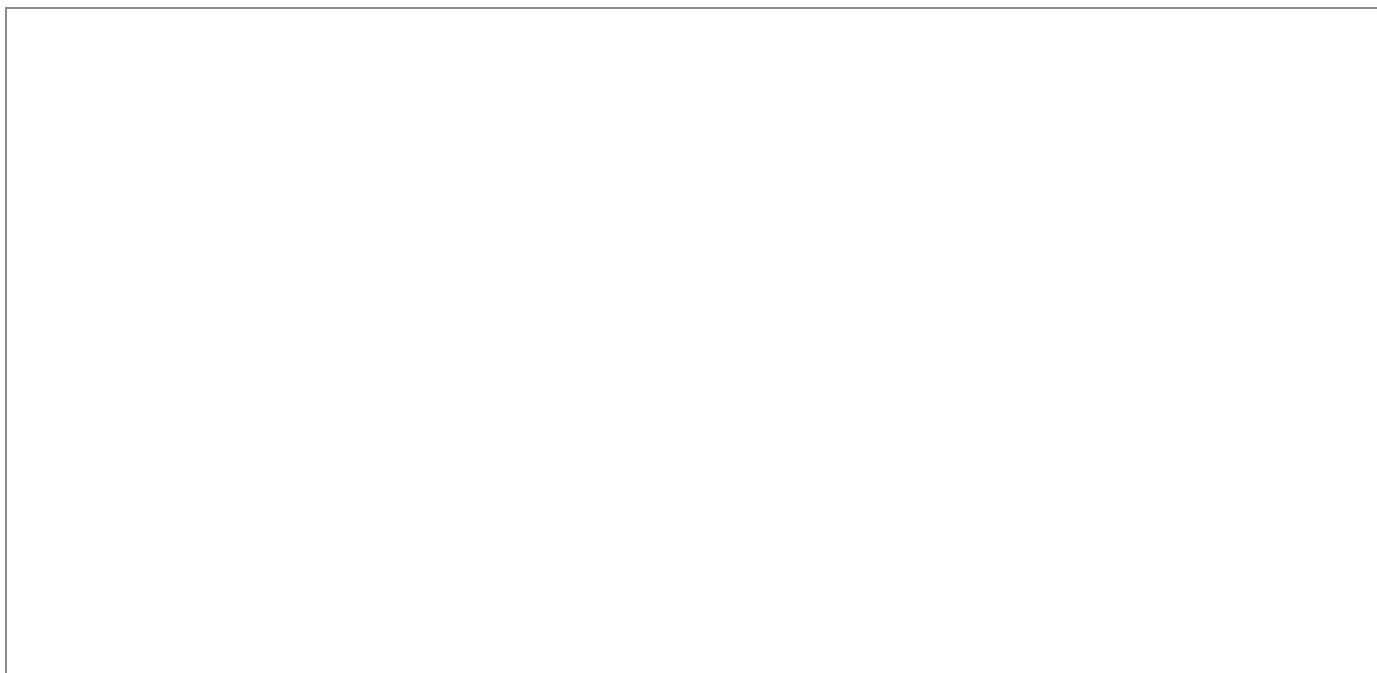
TGS details

Have the below been addressed on the TGS for this location?

Traffic volumes	☐ Yes	☐ No	☐ N/A	Details	
Predicted queue length	☐ Yes	☐ No	☐ N/A	Details	
Shoulder widths	☐ Yes	☐ No	☐ N/A	Details	
Sight distances	☐ Yes	☐ No	☐ N/A	Details	
Existing infrastructure	☐ Yes	☐ No	☐ N/A	Details	
Transport services	☐ Yes	☐ No	☐ N/A	Details	
Pedestrian generators	☐ Yes	☐ No	☐ N/A	Details	
Appropriate site access	☐ Yes	☐ No	☐ N/A	Details	
Appropriate escape route for traffic controllers	☐ Yes	☐ No	☐ N/A	Details	

Confirmation	
Does TGS require adjustments within tolerances? <i>If yes provide details TGS must include these adjustments with justification.</i>	
☐ Yes ☐ No	
Comments or details of action taken:	
Does TGS require any additional changes or modifications? <i>If yes provide details and return TGS to designer for additional changes or modifications</i>	
☐ Yes ☐ No	
Comments or details of action taken:	
Is TGS appropriate for use for works required at this location? <i>If no provide details and, return TGS into file and select alternative, if design returned to designer for correction</i>	
☐ Yes ☐ No	
Comments or details of action taken:	
Have key TTM risks been addressed on site? <i>If no, provide details and return TGS to designer for correction, review and approval</i>	
☐ Yes ☐ No	
Comments or details of action taken:	

Additional comments:



E.3 Weekly TTM inspection checklist

Weekly inspections must only be carried out by a PWZTMP qualified person. Weekly inspections must be carried out when a site is first open and at least once every week thereafter.

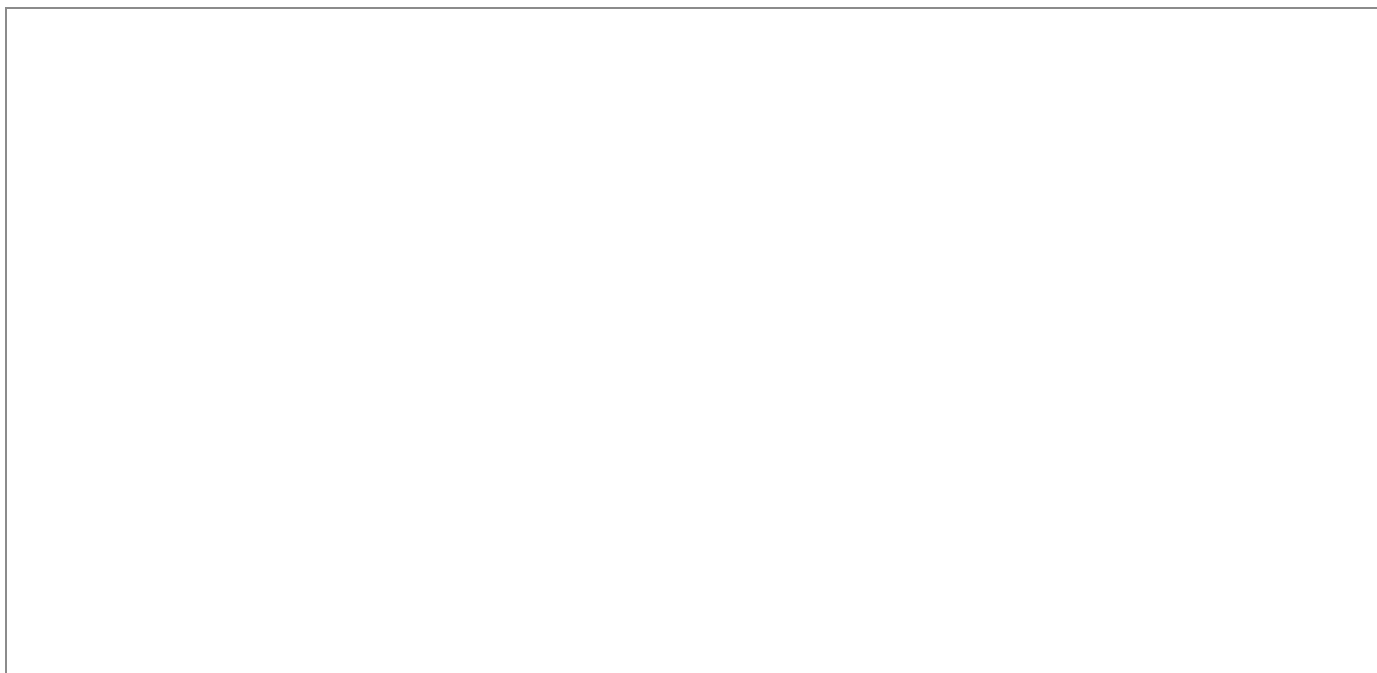
Completed by:			
Name:		Signature:	
TMP Reference:		TGS Reference:	
Date:		Inspection type	☐ Pre-opening ☐ Weekly
Desktop review			
Is a copy of the location TMP and relevant TGS available? <i>If no inspection must not be undertaken until documents are obtained</i>			☐ Yes ☐ No
Details of TMP and TGS:			
Are the location TMP and relevant TGS approved? <i>If no, work must be stopped until documents are approved</i>			☐ Yes ☐ No
Comments or details of action taken:			
Site Inspection			
Inspection completed:	☐ During the day ☐ During the night		
Signs and devices positioned as prescribed and commanding attention? <i>If no provide details and rectify signs</i>			☐ Yes ☐ No
Comments or details of action taken:			

Site Inspection		
Sign sizes as prescribed?		☐ Yes ☐ No <i>If no provide details and rectify signs</i>
Comments or details of action taken:		
Signs are mounted level and suitably clear of travel lanes?		☐ Yes ☐ No <i>If no provide details and rectify signs</i>
Comments or details of action taken:		
Has temporary delineation been applied as prescribed, with permanent markings obliterated?		☐ Yes ☐ No <i>If no provide details of action required to rectify delineation</i>
Comments or details of action taken:		
Are registered trailers i.e. VMS / light towers; suitably clear of travel lanes and delineated?		☐ Yes ☐ No <i>If no provide details and rectify location</i>
Comments or details of action taken:		
Are temporary speed zones operating as prescribed?		☐ Yes ☐ No <i>If no provide details and discuss with work supervisor</i>
Comments or details of action taken:		
Are PTCD positioned as prescribed in TGS?		☐ Yes ☐ No <i>If no provide details of action required to rectify</i>
Comments or details of action taken:		

Site Inspection		
Are manual traffic controllers clear of travel lane, have suitable escape route? <i>If no provide details of action required to rectify</i>		☐ Yes ☐ No
Comments or details of action taken:		
Are site accesses and egresses well defined and safe for work vehicles? <i>If no provide details of action required to rectify</i>		☐ Yes ☐ No
Comments or details of action taken:		
Termination signs are suitably located? i.e. D downstream of last activity. <i>If no provide details of action required to rectify</i>		☐ Yes ☐ No
Comments or details of action taken:		

Post site inspection confirmation	
Is worksite layout operating safely as intended? <i>If no provide details and implement controls to rectify</i>	
☐ Yes ☐ No	
Comments or details of action taken:	
Has TMP identified and addressed key TTM risks? <i>If no provide details and implement controls to rectify</i>	
☐ Yes ☐ No	
Comments or details of action taken:	
Have key TTM risks been addressed on site? <i>If no provide details of additional hazards and controls required</i>	
☐ Yes ☐ No	
Comments or details of action taken:	
Have copies of Shift Inspections been sighted as completed as required? <i>If no provide details and discuss with nominated rep completing Shift Inspections</i>	
☐ Yes ☐ No ☐ N/A	
Comments or details of action taken:	

Additional comments:



E.4 Shift/Daily TTM inspection checklist

Shift inspections must be undertaken by a person holding the PWZTMP or ITCP qualification when a TGS is installed, changed or updated, to ensure the TGS is implemented as designed. This includes at a minimum, twice per shift (recommended every 2 hours). This form can also be used for inspecting 'Aftercare' arrangements.

Completed by:					
Name:		Signature:			
TMP Reference:		TGS Reference:			
Date:		Time/s	Inspection 1	Inspection 2	Inspection 3
			00-00	00-00	00-00
Drive through TGS inspection					
Have any adjustments been made to the approved TGS?			Inspection 1	Inspection 2	Inspection 3
			☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
If yes, provide details:	Are changes within tolerances?		☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
	<i>If no, TGS must be reviewed by a PWZTMP</i>				
	Have changes been approved?		☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
	<i>If no, TGS must be approved</i>				
Comments or details of action taken:					
Have all signs and devices been installed in accordance with approved TGS?			☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
<i>If no, provide detail of action taken</i>					
Comments or details of action taken:					

Drive through TGS inspection		Inspection 1	Inspection 2	Inspection 3
Are PTCD positioned as prescribed in TGS? <i>If no, provide detail of action taken</i>		☐ Yes ☐ No ☐ N/A	☐ Yes ☐ No ☐ N/A	☐ Yes ☐ No ☐ N/A
Comments or details of action taken:				
Are manual traffic controllers clear of travel lane, have suitable escape route? <i>If no, provide detail and reposition manual traffic controllers</i>		☐ Yes ☐ No ☐ N/A	☐ Yes ☐ No ☐ N/A	☐ Yes ☐ No ☐ N/A
Comments or details of action taken:				
Are sign and devices in good condition, clearly visible to road users? <i>If no, provide detail of action taken</i>		☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Comments or details of action taken:				
Are all signs mounted level and suitably clear of travel lanes? <i>If no, provide detail of action taken</i>		☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Comments or details of action taken:				
Are conflicting or non-applicable signs covered or removed? <i>If no, provide detail and remove or cover signs</i>		☐ Yes ☐ No ☐ N/A	☐ Yes ☐ No ☐ N/A	☐ Yes ☐ No ☐ N/A
Comments or details of action taken:				

Drive through TGS inspection		Inspection 1	Inspection 2	Inspection 3
Is temporary delineation installed as prescribed i.e. straight line forming taper? <i>If no provide details and rectify delineation</i>		☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Comments or details of action taken:				
Have site conditions changed due to shade, park vehicles, glare etc. <i>If yes provide details and note if action is required</i>		☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Comments or details of action taken:				
Are registered trailers i.e. VMS / light towers; suitably clear of travel lanes and delineated? <i>If no provide details and rectify location</i>		☐ Yes ☐ No ☐ N/A	☐ Yes ☐ No ☐ N/A	☐ Yes ☐ No ☐ N/A
Comments or details of action taken:				
Are temporary speed zones operating as prescribed? <i>If no provide details and discuss with work supervisor</i>		☐ Yes ☐ No ☐ N/A	☐ Yes ☐ No ☐ N/A	☐ Yes ☐ No ☐ N/A
Comments or details of action taken:				
Are workers on foot / plant clearances been applied / observed? <i>If no provide details and implement controls to rectify</i>		☐ Yes ☐ No ☐ N/A	☐ Yes ☐ No ☐ N/A	☐ Yes ☐ No ☐ N/A
Comments or details of action taken:				

Post drive through confirmation		Inspection 1	Inspection 2	Inspection 3
Is TGS valid for the site activity and operating safely as intended? <i>If no provide details and implement controls to rectify</i>		☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Comments or details of action taken:				
Is TGS is appropriate for the current traffic conditions? <i>If no provide details and implement controls to rectify</i>		☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Comments or details of action taken:				
Have potential hazards identified in TGS been addressed? i.e. end-of-queue management <i>If no provide details of additional hazards and controls required</i>		☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Comments or details of action taken:				

Additional comments:

E.5 Post completion inspection checklist

Completed by:			
Name:		Road name/Staging Plan number:	
Signature:		Date / time:	
ITCP or PWZTMP card number			
Drive through post completed inspection			
Item		Comments / Action	
Have all work activities been completed?	☐ Yes ☐ No		
Has all plant and equipment been removed?	☐ Yes ☐ No		
Have all TTM signs and devices been removed?	☐ Yes ☐ No		
Has all TTM linemarking been obliterated?	☐ Yes ☐ No		
Have existing permanent speed limits been reinstated?	☐ Yes ☐ No		
Have all TTM site hazards been removed?	☐ Yes ☐ No		
Other	☐ Yes ☐ No		

Desktop post completion inspection		
Have all TGSs for completed tasks been retained?	☐ Yes ☐ No	
Have all TMP required documents been placed in relevant folders?	☐ Yes ☐ No	
Has TMP/TGS designer requested addition information post TTM removal?	☐ Yes ☐ No	
Is the road safe for opening to road users?	☐ Yes ☐ No	

Additional comments:

E.6 Portable variable message sign (VMS)/variable speed limit sign (VSLS) location and placement checklist

Proposed site location	
Road Details /Intersections /property numbers	
Purpose of the VMS/VSLS	
Proposed period of use	
Is this integrated with TMC?	

Checklist details for VMS/VSLS location	Yes, No or N/A	Comments/reasons for non-compliance
Planning		
Will the location of the proposed VMS/VSLS be in the road reserve?		
Will the proposed VMS/VSLS be visible from a road or road related area?		
Is the proposed VMS/VSLS being used as part of a major event?		
Safety		
Will the proposed location allow safe and easy access to the site for deployment of the portable VMS/VSLS?		
Is the proposed site located near any utilities (overhead or underground)?		
Will the proposed site cause personnel to be unsafely exposed to traffic?		
Will traffic control be required to safely place or remove the portable VMS/VSLS?		

Checklist details for VMS/VSLs location	Yes, No or N/A	Comments/reasons for non-compliance
Are there any other safety considerations at the proposed site? e.g. bore drains, culverts etc.		
Placement		
Is the proposed location likely to affect or change the patterns of any vulnerable road user movements?		
Is the proposed location likely to affect or change the pattern of cyclist movements?		
Will the proposed location be behind Transport approved safety barriers or as far away from the edge of the traffic lane as is practical in a position determined suitable based on a documented risk assessment and detailed in the TMP? For more information refer to Austroads <i>Guide to Road Design Part 6: Roadside Design, Safety and Barriers</i> .		
Is the proposed location at least 300m from the nearest permanent VMS?		
Is the proposed location at least 200-300m from significant static signs?		
Is the proposed location at least 200-300m from any signalised intersections?		
Will the proposed location cause driver distraction?		
Is the proposed location a suitable distance from any speed zoning signage?		
Is the proposed location in the direct run off carriageway path of a vehicle?		
Will the proposed location affect any residential or commercial properties?		

Checklist details for VMS/VSLs location	Yes, No or N/A	Comments/reasons for non-compliance
Will the proposed location affect any accesses or legal rights of way?		
Is the proposed location within 200m of any intersection or merging lane?		

Structures		
Will the proposed location be behind guard rail?		
Will the proposed location be behind wire rope fence?		
Is the proposed location close to significant road side furniture?		

THIS PAGE LEFT INTENTIONALLY BLANK

Contact Us:

If you have any questions or would like more information on this document please contact Transport for NSW:



roads-waterways.transport.nsw.gov.au



standards@transport.nsw.gov.au



13 22 13



Customer feedback
Locked Bag 928,
North Sydney NSW 2059



If you need help understanding this information, please contact the Translating and Interpreting Service on 131 450 and ask them to call us on 1800 131 782

February 2022
20.346